MATH 152
Mrs. Bonny Tighe

QUIZ 9
25 points
11.9 – 11.10

NAME Solutions

TA _____ Wed 4/17/2024

1. Find the derivative of the power series $f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{5^{k+1}(k+1)!}$

2pts

$$\sum_{k=1}^{\infty} \frac{(-1)^k \, 2k \, x^{2k-1}}{5^{k+1}(k+1)!}$$

can't be negative

2. Find the power series for the function $f(x) = \frac{x^3}{4+x^4}$ and find the interval of convergence.

$$\frac{\frac{x^3}{4}}{1-\left(-\frac{x^4}{4}\right)} = \sum_{n=0}^{\infty} \frac{x^3}{4}\left(-\frac{x^4}{4}\right)^n = \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+3}}{4^{n+1}}}$$

3pts

$$|r| = \left|\frac{x^4}{4}\right| < 1 \qquad |x^4| < 4 \qquad |x| < \sqrt{2}$$

$$I = (-\sqrt{2}, \sqrt{2})$$

3pts

3. Approximate the integral using power series and 4 non-zero terms.

$$\int_0^{0.2} \cos(2x^2)\,dx$$

$$\int_0^{0.2} \sum_{n=0}^{\infty} \frac{(-1)^n (2x^2)^{2n}}{(2n)!}\,dx = \int \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n} x^{4n}}{(2n)!}\,dx$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n} (x^{4n+1})}{(2n)!\,(4n+1)} \Bigg]_0^{0.2} = x - \frac{2^2 x^5}{2!(5)} + \frac{2^4 x^9}{4!(9)} - \frac{2^6 x^{13}}{6!(13)}$$

$$\left((0.2) - \frac{2^2(0.2)^5}{2!(5)} + \frac{2^4(0.2)^9}{4!(9)} - \frac{2^6(0.2)^{13}}{6!(13)} \right) - 0$$

4. Evaluate the limit using series $\lim_{x\to 0}\frac{\sin(x^2)-x^2}{x^6}$

3pts

$$\sin(x^2)=\sum_{n=0}^{\infty}\frac{(-1)^n(x^2)^{2n+1}}{(2n+1)!}=\sum_{n=0}^{\infty}\frac{(-1)^n x^{4n+2}}{(2n+1)!}=x^2-\frac{x^6}{3!}+\frac{x^{10}}{5!}-\frac{x^{14}}{7!}\cdots$$

$$\lim_{x\to 0}\frac{\cancel{x^2}-\frac{x^6}{3!}+\frac{x^{10}}{5!}-\frac{x^{14}}{7!}\cdots-\cancel{x^2}}{x^6}=\lim_{x\to 0}\left(\frac{-\frac{x^6}{3!}+\frac{x^{10}}{5!}-\frac{x^{14}}{7!}}{x^6}\right)$$

$$\lim_{x\to 0}\left(-\frac{1}{3!}+\frac{x^4}{5!}-\frac{x^8}{7!}\cdots\right)=-1/3!=-1/6$$

5. Use the Binomial Theorem to find the expansion of the function as a power series.

$$f(x)=\frac{1}{(1+2x^2)^2}=(1+2x^2)^{-2}\quad k=-2\qquad \sum_{n=0}^{\infty}\square\frac{(2x^2)^n}{n!}$$

$$\square=1+k+k(k-1)+k(k-1)(k-2)\cdots$$

$$=1-2-2(-3)-2(-3)(-4)-2(-3)(-4)(-5)$$

$$1-2!+3!-4!+5!\qquad (-1)^n(n+1)!$$

$n=0$ $n=1$

$$\sum_{n=0}^{\infty}\frac{(-1)^n(n+1)!}{n!}2^n x^{2n}$$

3pts

$$\boxed{\sum_{n=0}^{\infty}(-1)^n(n+1)2^n x^{2n}}$$

3pts

6. Find the sum of the power series $\sum_{k=0}^{\infty}\frac{(-1)^k\pi^{2k+1}}{6^{2k+1}(2k+1)!}$

$$=\sum_{k=0}^{\infty}\frac{(-1)^k(\pi/6)^{2k+1}}{(2k+1)!}$$

$\sin \pi/6 = 1/2$

7. Find the MacLaurin series using a known series

a) $f(x) = x^2 e^{3x^2}$

$$f(x) = x^2 \sum_{n=0}^{\infty} \frac{(3x^2)^n}{n!}$$

$$= \boxed{\sum_{n=0}^{\infty} \frac{3^n x^{2n+2}}{n!}}$$

2pts

b) $f(x) = \dfrac{\cos(3x^3)}{9}$

$$f(x) = \frac{1}{3^2} \sum_{n=0}^{\infty} \frac{(-1)^n (3x^3)^{2n}}{(2n)!}$$

$$= \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n-2} x^{6n}}{(2n)!}}$$

2pts

8. Find the Taylor series for $f(x) = \dfrac{1}{\sqrt{x+1}}$ centered at a = 8

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(8)(x-8)^n}{n!}$$

$f(x) = (x+1)^{-1/2}$

$f'(x) = -\frac{1}{2}(x+1)^{-3/2}$

$f''(x) = \frac{3}{4}(x+1)^{-5/2}$

$f'''(x) = \frac{-1\cdot3\cdot5}{2^3}(x+1)^{-7/2}$

$f''''(x) = \frac{1\cdot3\cdot5\cdot7}{2^4}(x+1)^{-9/2}$

$f(8) = (9)^{-1/2} = \frac{1}{3}$

$f'(8) = -\frac{1}{2}\cdot\frac{1}{3^3}$

$f''(8) = \frac{3}{2^2}\cdot\frac{1}{3^5}$

$f'''(8) = \frac{-1\cdot3\cdot5}{2^3}\cdot\frac{1}{3^7}$

$f''''(8) = \frac{1\cdot3\cdot5\cdot7}{2^4}\cdot\frac{1}{3^9}$

5pts

$$\underset{n=0}{\frac{1}{3}} - \underset{n=1}{\frac{1}{2\cdot3^2}} + \underset{n=2}{\frac{1\cdot3}{2^2 3^5}} - \underset{n=3}{\frac{1\cdot3\cdot5}{2^3 3^7}} + \underset{n=4}{\frac{1\cdot3\cdot5\cdot7}{2^4 3^9}}$$

$$f^{(n)}(8) = \frac{(-1)^n(1\cdot3\cdot5\cdots(2n-1))}{2^n 3^{2n+1}}$$ ← can it be negative

must leave n=0 outside Σ

$$\frac{1}{3} + \sum_{n=1}^{\infty} \frac{(-1)^n (1\cdot3\cdot5\cdots(2n-1))}{2^n 3^{2n+1}\, n!}(x-8)^n$$